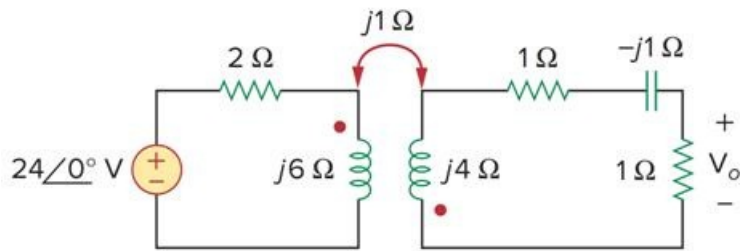


Problem

For the circuit in Fig. 13.76, find V_o .

Figure 13.76

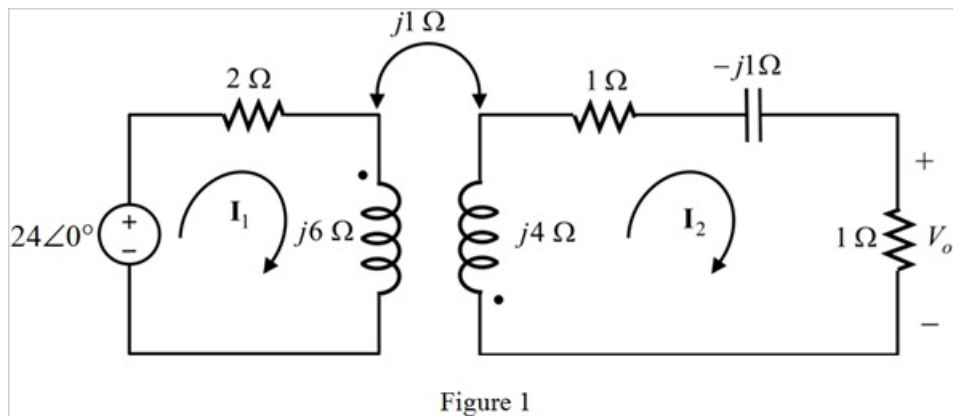


Step-by-step solution

Step 1 of 3

Refer to Figure 13.76 in the textbook for the circuit diagram.

Redraw the circuit with current loop directions is shown in Figure 1.



Step 2 of 3

Calculate the value of the current $\mathbf{I}_2$.

Apply Kirchhoff's Voltage Law in mesh-1.

$$(2 + j6)\mathbf{I}_1 + j\mathbf{I}_2 = 24\angle 0^\circ \dots (1)$$

Apply Kirchhoff's Voltage Law in mesh-2.

$$j\mathbf{I}_1 + (j4 + 1 - j + 1)\mathbf{I}_2 = 0$$

$$j\mathbf{I}_1 + (j3 + 2)\mathbf{I}_2 = 0$$

$$\mathbf{I}_1 = -\frac{(j3 + 2)}{j}\mathbf{I}_2$$

$$\mathbf{I}_1 = (-3 + j2)\mathbf{I}_2 \dots (2)$$

Substitute equation (2) in equation (1).

$$(2 + j6)((-3 + j2)\mathbf{I}_2) + j\mathbf{I}_2 = 24\angle 0^\circ$$

$$(-18 - j14 + j)\mathbf{I}_2 = 24\angle 0^\circ$$

$$\mathbf{I}_2 = \frac{24\angle 0^\circ}{(-18 - j13)}$$

Simplify $\mathbf{I}_2$ further.

$$\begin{aligned}\mathbf{I}_2 &= \frac{24\angle 0^\circ}{22.2\angle -144.16^\circ} \\ &= 1.081\angle 144.16^\circ \text{ A}\end{aligned}$$

Therefore, the value of the current $\mathbf{I}_2$ is $1.081\angle 144.16^\circ \text{ A}$.

Step 3 of 3

From the circuit, write the expression for the output voltage V_o .

$$V_o = \mathbf{I}_2(1)$$

Substitute $1.081\angle 144.16^\circ \text{ A}$ for $\mathbf{I}_2$.

$$V_o = 1.081\angle 144.16^\circ \text{ V}$$

Therefore, the value of the voltage V_o is $1.081\angle 144.16^\circ \text{ V}$.